Unit 5: A.C. Circuits

1. Fundamental A.C. Terminologies

Definition: Alternating Current (A.C.) is a current whose magnitude changes continuously and whose direction reverses periodically.

Key Definitions:

- **Cycle:** One complete set of positive and negative values of an alternating quantity.
- **Periodic Time (T):** The time taken by an alternating quantity to complete one full cycle. Measured in Seconds (s).
- **Frequency (f):** The number of cycles completed per second. $f = 1 / T$. Measured in Hertz (Hz).
- **Amplitude (Peak Value, V_m or I_m):** The maximum positive or negative value reached during a cycle.
- **Angular Velocity (ω):** The rate of change of phase angle. $\omega = 2\pi f$. Measured in rad/sec.

Important Values:

- **RMS Value (V_{rms}):** The effective value of A.C. that produces the same heating effect as an equivalent D.C. For sine wave: $V_{rms} = \frac{V_m}{\sqrt{2}} = 0.707 V_m$.
- **Average Value (V_{avg}):** The mean of all instantaneous values over a half-cycle. For sine wave: $V_{avg} = \frac{2V_m}{\pi} = 0.637 V_m$.
- **Form Factor:** Ratio of RMS value to Average value. For a pure sine wave, Form Factor = 1.11.
- **Peak Factor:** Ratio of Peak value to RMS value. For a pure sine wave, Peak Factor = $\sqrt{2} = 1.414$.

👉 **Hinglish Explanation:** AC signal har pal badalta hai aur apni direction reverse karta hai. Ek pura round (positive + negative) ek **Cycle** kehlata hai. Ek second me jitne cycles pure honge, wo uski **Frequency** (Hz) hai (India me 50Hz hai). **RMS value** wo DC equivalent hai jo same heat paida karta hai. **Form Factor (1.11)** aur **Peak Factor (1.414)** exams me direct objective type me puche jate hain!

2. Mathematical & Phasor Representation

Definition: An alternating quantity can be represented mathematically by trigonometric equations and visually by rotating vectors called phasors.

Mathematical Representation:

The instantaneous voltage (v) and current (i) at any time t are given by:

$$v = V_m \sin(\omega t)$$
$$i = I_m \sin(\omega t \pm \phi)$$

Where ϕ is the **Phase Angle**. It represents the angular difference between voltage and current. If it is positive (+), current leads voltage. If negative (-), current lags voltage.

Phasor Diagram:

A phasor is a straight line (vector) rotating counter-clockwise at an angular velocity ω. The length of the line represents the maximum value (Amplitude) or RMS value, and its angle with the reference axis represents the phase.

> 👉 **Hinglish Explanation:** AC wave ko math ke formulas se dikhane ke liye hum sine function ka use karte hain ($V_m \sin \omega t$). Voltage aur current hamesha ek sath chalein aisa zaroori nahi hai. Agar ek aage aur ek peeche ho, toh unke beech ke angle ko **Phase Angle (ϕ)** kehte hain. **Phasor diagram** inhe lines aur angles ke roop me paper par draw karne ka aasan tareeka hai.

3. A.C. through Pure R, L, and C Circuits

Definition: The behavior of an AC circuit depends entirely on the type of component (Resistor, Inductor, or Capacitor) connected to it. They affect the phase relationship between voltage and current.

Component	Phase Relationship	Opposition to Current	Formula
Pure Resistor (R)	Voltage and Current are **in phase** ($\phi = 0^\circ$).	Resistance (R)	$R = V / I$
Pure Inductor (L)	Current **lags** Voltage by 90° ($\phi = 90^\circ$).	Inductive Reactance (X_L)	$X_L = \omega L = 2\pi f L$
Pure Capacitor (C)	Current **leads** Voltage by 90° ($\phi = 90^\circ$).	Capacitive Reactance (X_C)	$X_C = \frac{1}{\omega C} = \frac{1}{2\pi f C}$

Impedance (Z):

The total opposition offered by an AC circuit to the flow of current. It is a combination of Resistance (R) and Reactance (X). Measured in Ohms (Ω).

> 👉 **Hinglish Explanation:** Isme sabse zaroori yaad rakhne ka trick hai **CIVIL**.
> **C I V** -> Capacitor (C) me Current (I) aage hota hai Voltage (V) se.
> **V I L** -> Voltage (V) aage hota hai Current (I) se Inductor (L) me.

Resistor ekdam seedha saadha hai, usme V aur I dono ek sath chalte hain (0 degree gap). Inductor ka opposition frequency badhne par badhta hai, aur Capacitor ka opposition frequency badhne par kam hota hai.

4. Series A.C. Circuits (R-L, R-C, R-L-C)

Definition: In real-world circuits, components are not purely resistive, inductive, or capacitive. They are connected in combinations (Series or Parallel) creating total Impedance (Z).

1. R-L Series Circuit:

A resistor and inductor in series. Current lags voltage by an angle ϕ (between 0° and 90°).
Impedance: $Z = \sqrt{R^2 + X_L^2}$
Phase Angle: $\tan \phi = \frac{X_L}{R}$

2. R-C Series Circuit:

A resistor and capacitor in series. Current leads voltage by an angle ϕ (between 0° and 90°).
Impedance: $Z = \sqrt{R^2 + X_C^2}$
Phase Angle: $\tan \phi = \frac{X_C}{R}$

3. R-L-C Series Circuit:

A resistor, inductor, and capacitor all connected in series. The overall behavior depends on whether $X_L > X_C$ (inductive) or $X_C > X_L$ (capacitive).

$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$

Series Resonance:

A special condition occurs when $X_L = X_C$. The reactive parts cancel each other out. Total impedance becomes minimum ($Z = R$), and current becomes maximum. This is called **Resonance**.

👉 **Hinglish Explanation:** Agar Resistor aur Inductor dono line me lage hon (R-L series), to current voltage se peeche (lag) hota hai par pure 90 degree se nahi. Is combination ke total rokne wale power ko **Impedance (Z)** bolte hain. R-L-C circuit me sabse interesting cheez tab hoti hai jab Inductor ka X_L aur Capacitor ka X_C barabar ho jate hain. Ye dono ek dusre ko cancel kar dete hain aur circuit bilkul pure Resistor jaisa behave karta hai. Ise **Resonance** kehte hain.

5. Power in A.C. Circuits & Power Triangle

Definition: Unlike DC circuits, power in AC circuits is divided into three components due to the phase difference between voltage and current. These are Active, Reactive, and Apparent power.

Types of Power:

- **Active / Real / True Power (P):** The actual power consumed by the resistive part of the circuit to do useful work.

Formula: $P = VI \cos\phi$. Unit: **Watts (W)** or kW.

- **Reactive Power (Q):** The power that continuously bounces back and forth between the source and the reactive components (L or C). It does no useful work.
  Formula: $Q = VI \sin\phi$. Unit: **Volt-Ampere Reactive (VAR)** or kVAR.
- **Apparent Power (S):** The total power supplied by the source. It is the vector sum of Active and Reactive power.
  Formula: $S = VI$. Unit: **Volt-Amperes (VA)** or kVA.

Power Factor ($\cos\phi$):

The ratio of Active Power to Apparent Power (P/S). It indicates how efficiently electrical power is being used.
Power Factor = $\cos\phi = \frac{R}{Z}$. Its value ranges from 0 to 1.

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POWER TRIANGLE:

           /|
     (S)  / |
     VA  /  | (Q) VAR
        /   | Reactive Power
       /    |
      /phi__|
      (P) Watts
     Active Power

      S^2 = P^2 + Q^2
```

👉 **Hinglish Explanation:** AC me jo power supply aati hai use Apparent power (VA) bolte hain. Jisme se jo actual me bulb/fan ko chalane me kaam aati hai wo Active power (Watt) hai, aur jo Inductor/Capacitor me waste hokar wapas chali jati hai wo Reactive power (VAR) hai. In teeno ka relation Pythagoras theorem jaise **Power Triangle** se dikhaya jata hai. **Power Factor ($\cos\phi$)** hamesha 1 ke kareeb hona chahiye taaki power waste na ho.

6. Three-Phase Systems: Star and Delta Connections

Definition: A 3-phase system generates, transmits, and distributes AC power using three separate voltages that are out of phase with each other by 120 degrees. The coils can be connected in Star (Y) or Delta (Δ) configurations.

1. Star (Y) Connection:

One end of all three coils is connected to a common central point called the **Neutral point**. The other ends form the three Line phases (R, Y, B).

- **Line Current (I_L) vs Phase Current (I_{ph}):** $I_L = I_{ph}$
- **Line Voltage (V_L) vs Phase Voltage (V_{ph}):** $V_L = \sqrt{3} \times V_{ph}$
- **Advantage:** Provides a neutral wire, giving two voltage levels (e.g., 400V Line-to-Line and 230V Line-to-Neutral). Good for lighting and mixed loads.

2. Delta (Δ) Connection:

The end of one coil is connected to the starting end of the next coil, forming a closed loop (triangle). There is NO neutral point.

- **Line Voltage (V_L) vs Phase Voltage (V_{ph}):** $V_L = V_{ph}$
- **Line Current (I_L) vs Phase Current (I_{ph}):** $I_L = \sqrt{3} \times I_{ph}$
- **Advantage:** Ideal for transmitting power and running heavy 3-phase induction motors.

Total 3-Phase Power:

$$P_{total} = \sqrt{3} \times V_L \times I_L \times \cos\phi$$

👉 **Hinglish Explanation:** Gharon me single-phase aata hai, par transmission aur industry me **3-Phase** power use hota hai jisme 3 wires hote hain (R, Y, B), aur sab 120 degree ke gap par hote hain. **Star connection** (Y shape) me beech se ek 4th wire (Neutral) nikalta hai, isme Line voltage Phase voltage ka $\sqrt{3}$ guna bada hota hai. **Delta connection** (Triangle shape) me neutral nahi hota, waha voltage barabar hoti hai par Line current Phase current se $\sqrt{3}$ guna bada hota hai.

🔁 Quick Revision Notes (Exam Saver)

- **RMS Value:** $V_{rms} = V_m / \sqrt{2}$. Average Value: $V_{avg} = 2V_m / \pi$. Form factor = 1.11.
- **Pure Components:** R (In phase, $\phi=0$). L (I lags V by 90°). C (I leads V by 90°). Memory trick: CIVIL.
- **Impedance (Z):** Total opposition. $Z = \sqrt{R^2 + (X_L - X_C)^2}$. Resonance occurs when $X_L = X_C$.
- **Power Triangle:** Active (W) = $VI\cos\phi$. Reactive (VAR) = $VI\sin\phi$. Apparent (VA) = VI.
- **Power Factor:** $\cos\phi = R / Z$. Efficiency of power transfer.
- **Star Connection:** $I_L = I_{ph}$, $V_L = \sqrt{3} V_{ph}$. (Has Neutral).
- **Delta Connection:** $V_L = V_{ph}$, $I_L = \sqrt{3} I_{ph}$. (No Neutral).

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